Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

ANSWER KEY

Section 1: Domains and Ranges of Inverse Trig Functions

1 What are the domain and range of the principal arcsine function $y = \arcsin x$?

a. **Domain:** $[-1, 1]$**; Range:** $[-\frac{\pi}{2}, \frac{\pi}{2}]$ ✓

b. Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

c. Domain: $[-1, 1]$; Range: $[0, \pi]$

d. Domain: $(-\infty, \infty)$; Range: $(-\frac{\pi}{2}, \frac{\pi}{2})$

To make sine invertible, its domain is restricted to $[-\pi/2, \pi/2]$ with range $[-1, 1]$. Thus $\arcsin x$ has domain $[-1, 1]$ and range $[-\pi/2, \pi/2]$.

2 What are the domain and range of $y = \arccos x$?

a. **Domain:** $[-1, 1]$**; Range:** $[0, \pi]$ ✓

b. Domain: $[-1, 1]$; Range: $[-\frac{\pi}{2}, \frac{\pi}{2}]$

c. Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

d. Domain: $[0, 1]$; Range: $[0, \pi/2]$

The cosine function is restricted to $[0, \pi]$ where it is strictly decreasing. Therefore $\arccos x$ maps $[-1, 1]$ onto $[0, \pi]$.

3 What are the horizontal asymptotes of the graph of $y = \arctan x$?

a. $y = \frac{\pi}{2}$ **and** $y = -\frac{\pi}{2}$ ✓

b. $y = \pi$ and $y = 0$

c. $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$

d. $y = 1$ and $y = -1$

As $x \to \infty$, $\arctan x \to \frac{\pi}{2}$; as $x \to -\infty$, $\arctan x \to -\frac{\pi}{2}$.

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Section 2: Derivatives of Inverse Trigonometric Functions

4 What is the derivative of $f(x) = \arcsin(3x)$ for $|x| < \frac{1}{3}$?

a. $\frac{3}{\sqrt{1-9x^2}}$ ✓

b. $\frac{1}{\sqrt{1-9x^2}}$

c. $\frac{3}{\sqrt{1-3x^2}}$

d. $\frac{3}{1+9x^2}$

By the chain rule: $\frac{d}{dx}[\arcsin(u)] = \frac{u'}{\sqrt{1-u^2}} = \frac{3}{\sqrt{1-(3x)^2}} = \frac{3}{\sqrt{1-9x^2}}$.

5 Find the derivative of $g(x) = \arctan\left(\frac{x}{2}\right)$.

a. $\frac{2}{4+x^2}$ ✓

b. $\frac{1}{1+x^2/4}$

c. $\frac{1}{4+x^2}$

d. $\frac{2}{\sqrt{4-x^2}}$

$\frac{d}{dx}[\arctan(x/2)] = \frac{1/2}{1+(x/2)^2} = \frac{1/2}{1+x^2/4} = \frac{1/2}{(4+x^2)/4} = \frac{2}{4+x^2}$.

6 What is the relationship between the derivatives of $\arcsin x$ and $\arccos x$?

a. $\frac{d}{dx}[\arccos x] = -\frac{d}{dx}[\arcsin x] = -\frac{1}{\sqrt{1-x^2}}$ ✓

b. $\frac{d}{dx}[\arccos x] = \frac{d}{dx}[\arcsin x]$

c. $\frac{d}{dx}[\arccos x] = \frac{1}{1-x^2}$

d. They are reciprocals of each other

Because $\arcsin x + \arccos x = \frac{\pi}{2}$ for all $x \in [-1, 1]$, differentiating both sides yields $\frac{d}{dx}[\arccos x] = -\frac{d}{dx}[\arcsin x]$.

Section 3: Tangent Lines and Extreme Values

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7 Find the slope of the tangent line to $y = \arcsin x$ at $x = 0$.

a. 1 ✓

b. 0

c. $\pi/2$

d. Undefined

$y'(x) = \frac{1}{\sqrt{1-x^2}}$. At $x = 0$, $y'(0) = \frac{1}{\sqrt{1-0}} = 1$.

8 What happens to the slope of $y = \arcsin x$ as $x \to 1^-$?

a. **The slope approaches $+\infty$ (vertical tangent)** ✓

b. The slope approaches 0

c. The slope approaches 1

d. The slope approaches -1

As $x \to 1^-$, $\sqrt{1-x^2} \to 0^+$, so $y'(x) = \frac{1}{\sqrt{1-x^2}} \to +\infty$, resulting in a vertical tangent at the boundary.

9 Find the equation of the tangent line to $y = \arctan x$ at $x = 1$.

a. $y - \frac{\pi}{4} = \frac{1}{2}(x-1)$ ✓

b. $y - \frac{\pi}{4} = 1(x-1)$

c. $y = \frac{1}{2}x + \frac{\pi}{4}$

d. $y - 1 = \frac{1}{2}(x - \frac{\pi}{4})$

Point is $(1, \arctan 1) = (1, \pi/4)$. Derivative is $y'(1) = \frac{1}{1+1^2} = \frac{1}{2}$. Tangent line: $y - \frac{\pi}{4} = \frac{1}{2}(x-1)$.

Section 4: Integration Producing Inverse Trig Functions

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10 Evaluate $\int_0^{1/2} \frac{1}{\sqrt{1-x^2}}dx$.

a. $\frac{\pi}{6}$ ✓

b. $\frac{\pi}{3}$

c. $\frac{\pi}{4}$

d. $\frac{1}{2}$

$[\arcsin x]_0^{1/2} = \arcsin(1/2) - \arcsin(0) = \frac{\pi}{6} - 0 = \frac{\pi}{6}$.

11 Evaluate $\int_0^1 \frac{1}{1+x^2}dx$.

a. $\frac{\pi}{4}$ ✓

b. $\frac{\pi}{2}$

c. 1

d. $\ln 2$

$[\arctan x]_0^1 = \arctan(1) - \arctan(0) = \frac{\pi}{4} - 0 = \frac{\pi}{4}$.

12 Evaluate $\int \frac{1}{\sqrt{4-9x^2}}dx$.

a. $\frac{1}{3}\arcsin\left(\frac{3x}{2}\right) + C$ ✓

b. $\frac{1}{2}\arcsin\left(\frac{3x}{2}\right) + C$

c. $\arcsin\left(\frac{3x}{2}\right) + C$

d. $\frac{1}{6}\arcsin\left(\frac{3x}{2}\right) + C$

Factor out 4: $\sqrt{4(1-9x^2/4)} = 2\sqrt{1-(3x/2)^2}$. Let $u = 3x/2$, $du = 3/2dx \Rightarrow dx = 2/3du$. Integral: $\frac{1}{2} \cdot \frac{2}{3}\int \frac{du}{\sqrt{1-u^2}} = \frac{1}{3}\arcsin(u) + C = \frac{1}{3}\arcsin(3x/2) + C$.

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